Harrison Guerrero

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PROFILE

Full Stack AI Engineer with hands-on experience designing and deploying intelligent, data-driven applications. Skilled in building scalable backends (Python, Node.js) and dynamic frontends (React, Next.js) with seamless AI integrations using modern LLM and RAG frameworks. Strong background in cloud architecture (AWS, GCP) and real-time systems, bridging software engineering with applied machine learning.

PROFESSIONAL EXPERIENCE

Nooks AI — 07/2022 – 08/2025 | San Francisco, CA

Senior Software Engineer - Core Platform

- Architected and scaled AI-native sales automation systems supporting **thousands of concurrent real-time calls**, processing multimodal audio/text streams with less 300ms inference latency.
- Designed and implemented **LLM** orchestration pipelines using **OpenAI APIs, LangChain, and vector search (Pinecone/FAISS)** for real-time summarization, objection detection, intent classification, and CRM enrichment.
- Built a high-throughput backend platform (**Node.js** + **Python** microservices) using async processing, **WebSockets**, **Redis**, and event-driven messaging to support streaming AI inference workflows.
- Led platform-level API contract design and schema versioning strategy, enabling internal teams to ship AI features independently without breaking backward compatibility.
- Developed **RAG**-based knowledge grounding systems that improved summary accuracy by 28% and reduced hallucinations in enterprise deployments.
- Implemented evaluation pipelines (golden datasets + automated scoring) to benchmark prompt quality, token efficiency, and response reliability.
- Reduced **LLM** operating cost by 35% via dynamic prompt routing, caching embeddings, and context-window optimization.
- Mentored 4+ engineers on distributed system design, AI reliability patterns, and scalable frontend architecture (**React** + **Zustand**).
- Partnered closely with product leadership to ship predictive lead scoring and AI copilot features that contributed to **10× customer growth** over three years.

Aretum, LLC — 02/2020 – 08/2022 | Washington DC

Software Engineer

- Led development of *FedRequest*, an **AI-powered FOIA request analysis platform**, automating document ingestion, redaction, and classification pipelines to reduce manual review time by 60%.
- Architected full-stack microservice solutions using **Python (FastAPI), Node.js, React,** and **PostgreSQL**, ensuring scalability, modularity, and maintainability across multiple Aretum products.
- Implemented DevOps pipelines with **Docker**, **Kubernetes**, and **AWS (EKS, Lambda, RDS)**, achieving zero-downtime deployments and streamlined CI/CD via GitHub Actions and Terraform.
- Established secure data handling workflows aligned with FedRAMP and NIST 800-53 standards, incorporating encryption-at-rest, IAM roles, and automated audit logging for compliance.
- Optimized system scalability through horizontal scaling, Redis caching, and asynchronous background tasks (Celery, SQS), enabling the platform to handle thousands of FOIA requests in parallel.
- Integrated advanced monitoring and observability using **Prometheus**, **Grafana**, and **CloudWatch**, improving issue detection time and infrastructure reliability.
- Collaborated with cross-functional federal partners to deliver compliant, production-ready solutions under strict SLAs and data governance constraints.

Omada Health 04/2018 – 02/2020 | San Francisco, CA

Full Stack Developer

- Led development of a **remote patient monitoring (RPM) system** that integrated **IoT-connected health devices** (blood pressure cuffs, glucose monitors, weight scales) with the Omada platform for real-time patient data tracking.
- Architected secure data pipelines using **Python (FastAPI), Node.js, and PostgreSQL**, ensuring **HIPAA and GDPR compliance** through encryption, access control, and PHI anonymization.
- Designed scalable microservices to process high-frequency health telemetry data, enabling proactive alerts and clinician dashboards with sub-second data synchronization.
- Collaborated with clinicians, data scientists, and compliance teams to implement patient consent workflows and ensure end-to-end traceability of sensitive health data.
- Built analytics dashboards using **React, TypeScript,** and **D3.js**, visualizing key patient metrics and trends to support early intervention and personalized care.

Microsoft

Software Engineer Intern

- Built an internal telemetry dashboard for Azure engineering teams using **C#, .NET Core, and React**, visualizing live service health metrics and deployment trends across regional clusters.
- Developed microservices to aggregate and normalize telemetry data from Azure Monitor and Application Insights, improving diagnostic visibility and reducing incident triage time by 40%.

SKILLS

AI & Machine Learning

LLM & RAG pipelines (OpenAI, LangChain, DSPy), TensorFlow/PyTorch integration, NLP & CV model serving (FastAPI, Flask), vector DBs (Pinecone, FAISS), prompt engineering, AI agent frameworks

Backend Development

Python (FastAPI, Flask), Node.js (Express, Nest.js), C#, GraphQL/REST APIs, real-time systems (WebSockets), microservices & async processing (Celery, RQ)

Frontend Engineering

React, Next.js, TypeScript, Redux, Zustand, Material UI, TailwindCSS, D3.js/Recharts, AI-driven dashboards & copilot UIs

Data & Visualization

PostgreSQL, MongoDB, Redis, Supabase, BigQuery, ETL pipelines, time-series analytics

DevOps & Cloud Infrastructure

Docker, Kubernetes, AWS (EKS, Lambda, RDS, S3), GCP, CI/CD (Jenkins, GitHub Actions), serverless AI deployment, Cloudflare Workers

Performance & Tooling

Profiling & optimization (Lighthouse, Chrome Tracing), Webpack/Vite bundling, Git workflows, automated testing (Jest, Playwright)

EDUCATION

Chapman University 2013 – 2017 | Orange, CA

Bachelor's degree in Computer Engineering